

“Calculus 3”

Multi-Variable Calculus

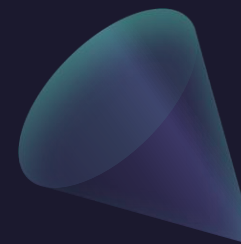
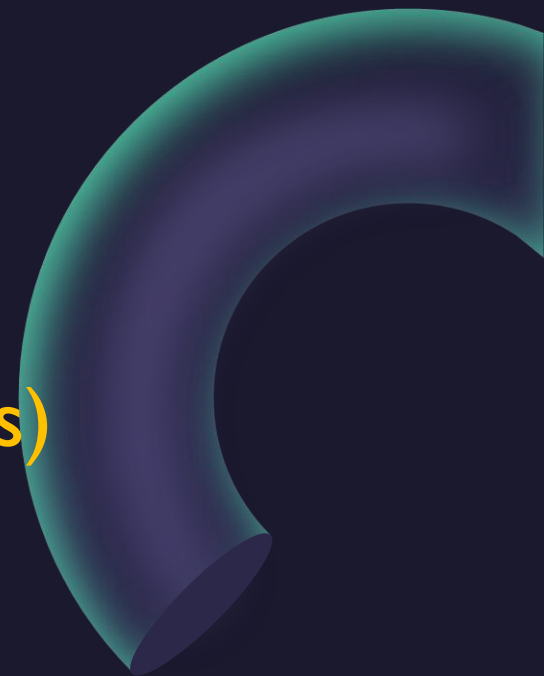
Instructor: Álvaro Lozano-Robledo

Day 20



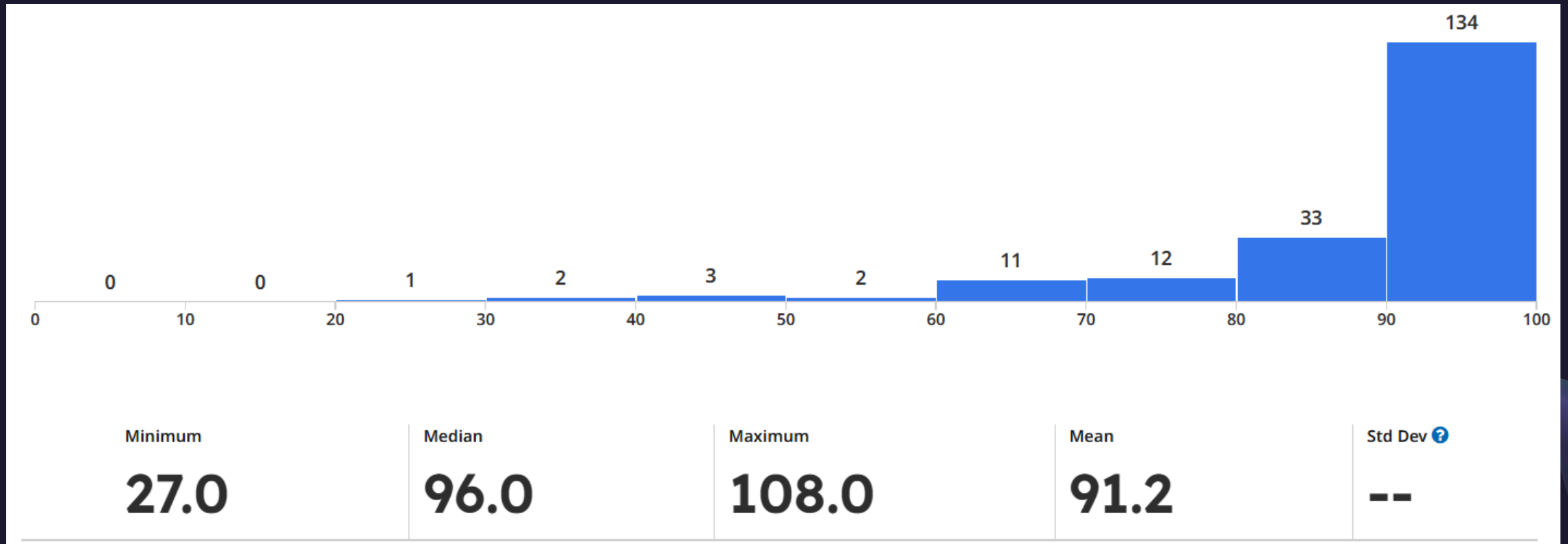
Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.2 (line integrals)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



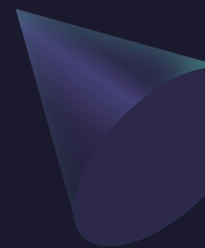


Exam 2: Stats!





ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

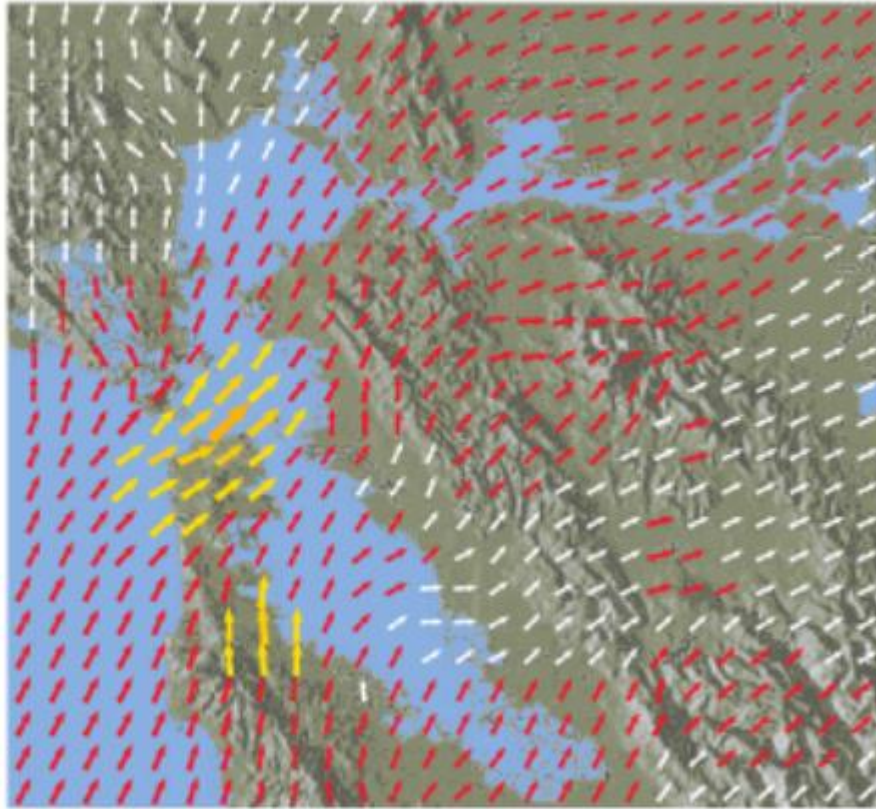
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Vector Fields

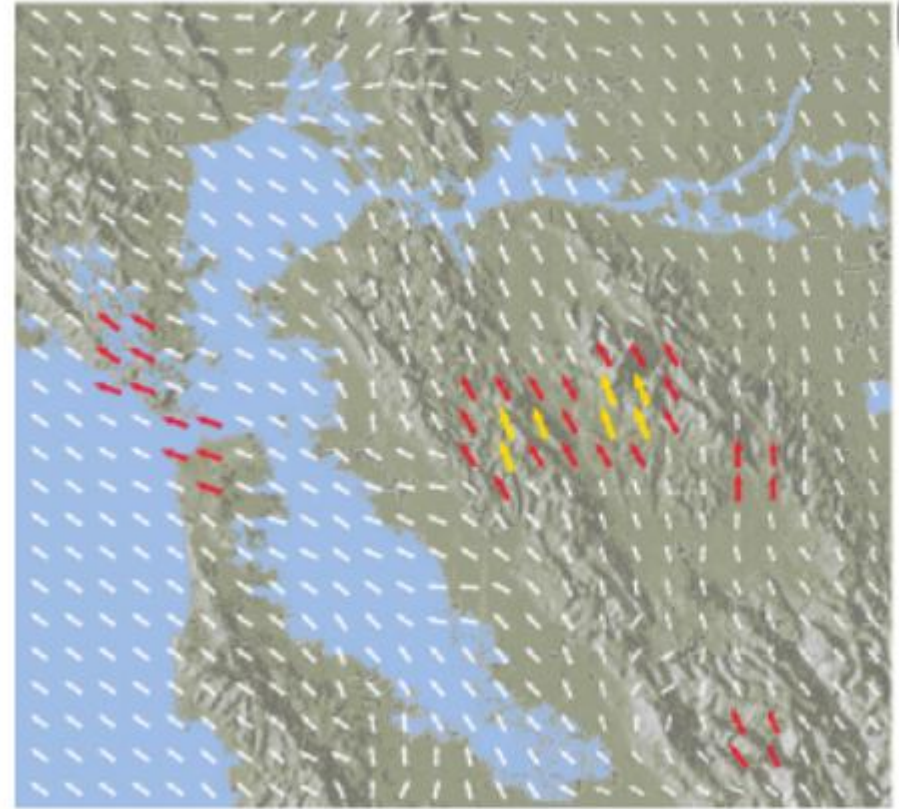


Vector Fields

Velocity vector fields showing San Francisco Bay wind patterns on a particular spring day



(a) 6:00 PM



(b) 6:00 AM

Vector Fields

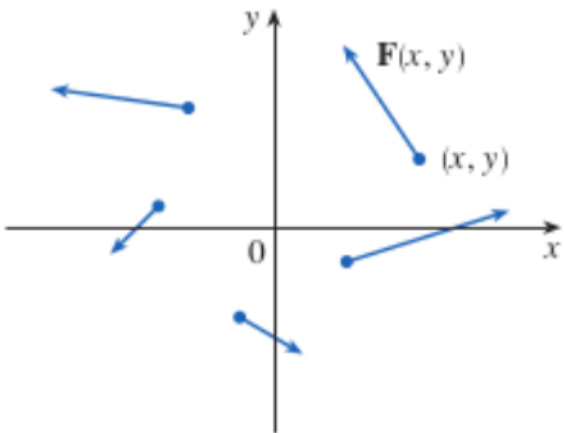


Vector Fields

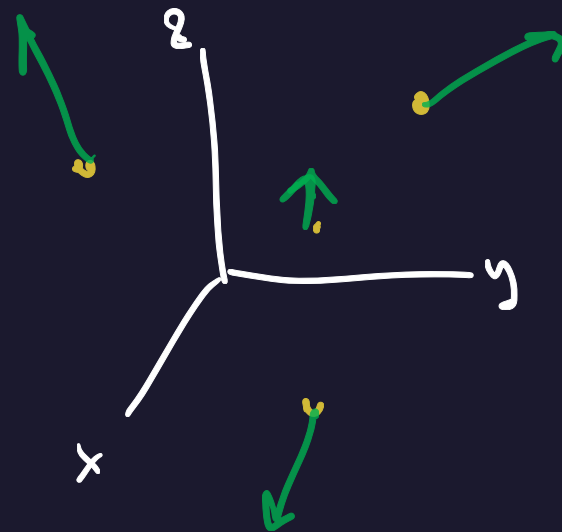
$$\begin{aligned} \mathbf{F}: \mathbb{R}^2 &\longrightarrow \mathbb{R}^2 \\ (x, y) &\longmapsto \checkmark \end{aligned}$$

Let D be a set in $\mathbb{R}^2$ (a plane region). A **vector field on $\mathbb{R}^2$** is a function $\mathbf{F}$ that assigns to each point (x, y) in D a two-dimensional vector $\mathbf{F}(x, y)$.

Let E be a subset of $\mathbb{R}^3$. A **vector field on $\mathbb{R}^3$** is a function $\mathbf{F}$ that assigns to each point (x, y, z) in E a three-dimensional vector $\mathbf{F}(x, y, z)$.



$$\begin{aligned} \mathbf{F}: \mathbb{R}^3 &\longrightarrow \mathbb{R}^3 \\ P &\longmapsto \mathbf{v}_P \end{aligned}$$

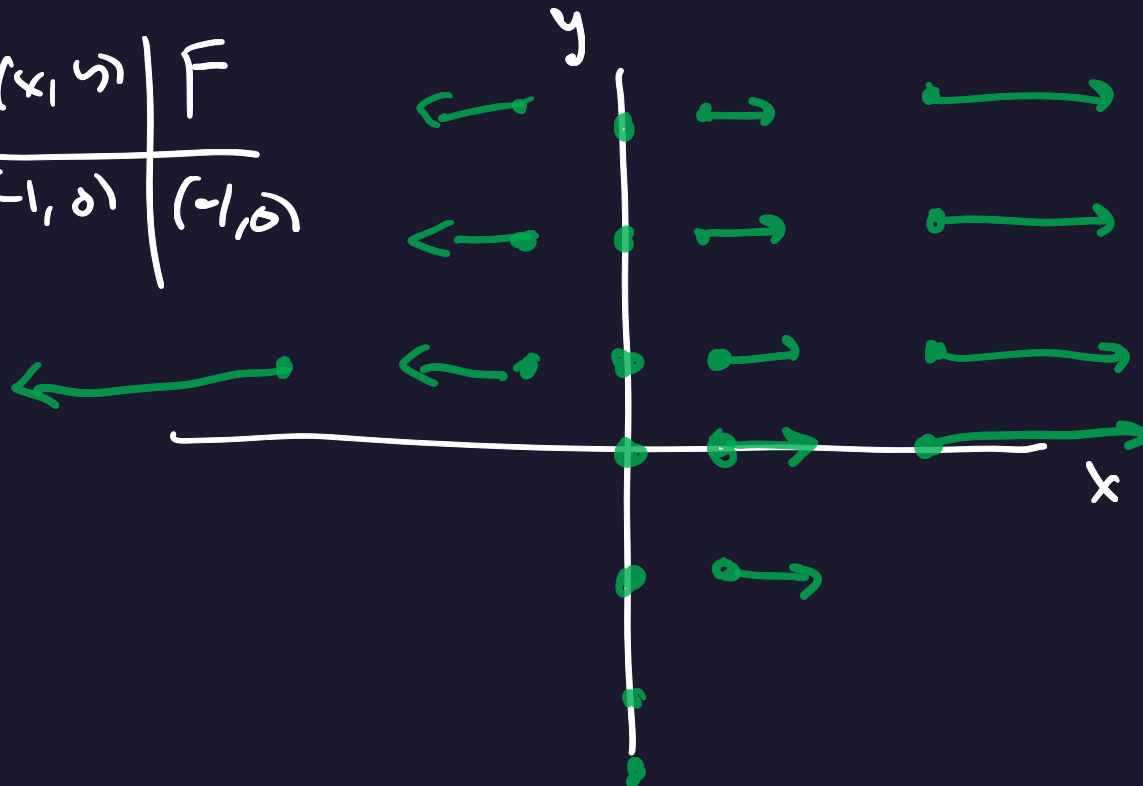


Example: Sketch a vector field on the plane given by

$$F(x, y) = (x, 0) = x \cdot i + 0 \cdot j$$

(x, y)	$F(x, y)$
$(0, 0)$	$(0, 0)$
$(1, 0)$	$(1, 0)$
$(2, 0)$	$(2, 0)$
$(0, 1)$	$(0, 0)$
$(0, 2)$	$(0, 0)$
$(1, 1)$	$(1, 0)$
$(1, 2)$	$(1, 0)$

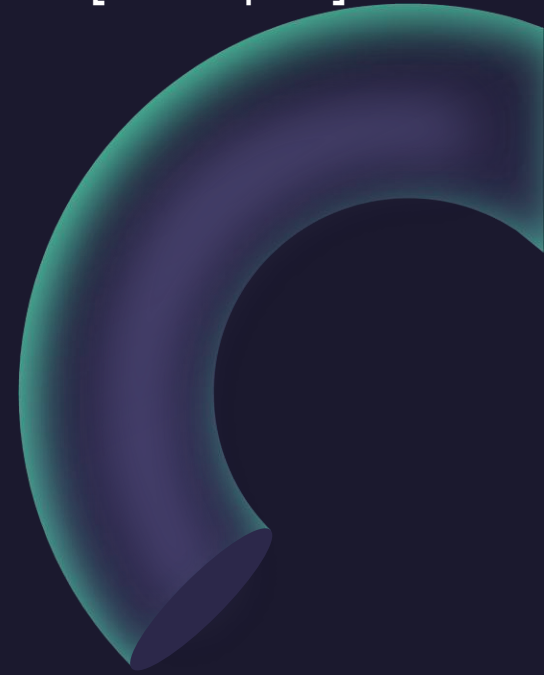
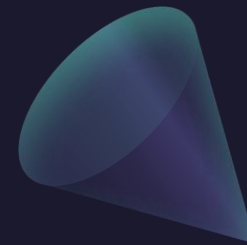
(x, y)	F
$(-1, 0)$	$(-1, 0)$



$$|F(x, y)| = |(x, 0)| = \sqrt{x^2 + 0^2} = |x|$$

Example: Sketch a vector field on the plane given by

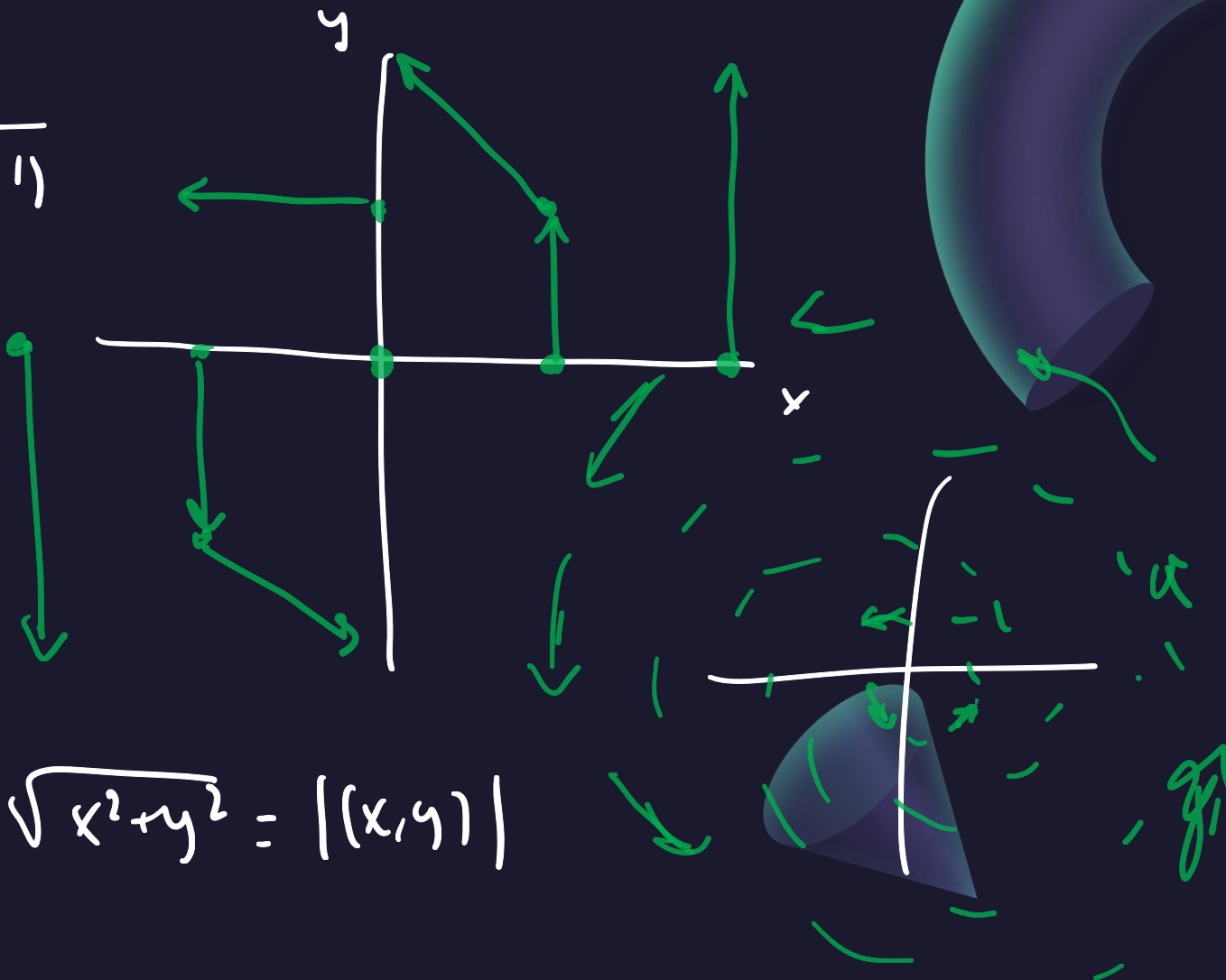
$$F(x, y) = (x, 0) = x \cdot i + 0 \cdot j$$



Example: Sketch a vector field on the plane given by

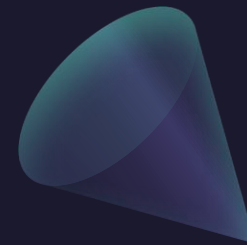
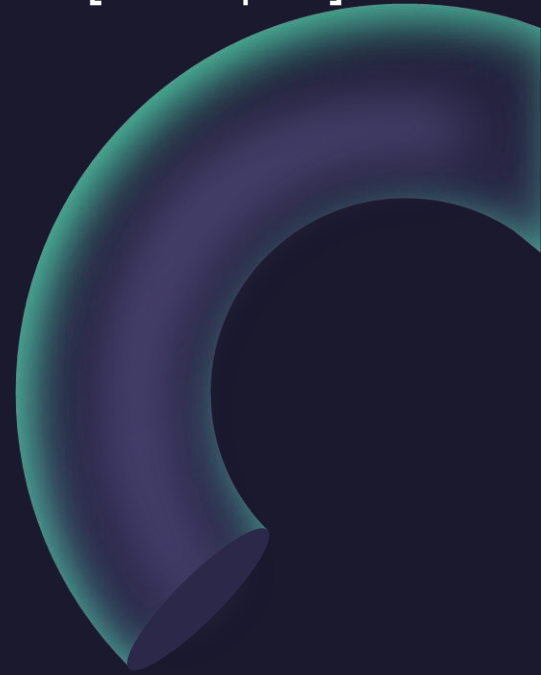
$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$

(x, y)	F	(x, y)	F
$(0, 0)$	$(0, 0)$	$(1, 1)$	$(-1, 1)$
$(1, 0)$	$(0, 1)$		
$(0, 1)$	$(-1, 0)$		
$(-1, 0)$	$(0, -1)$		
$(2, 0)$	$(0, 2)$		

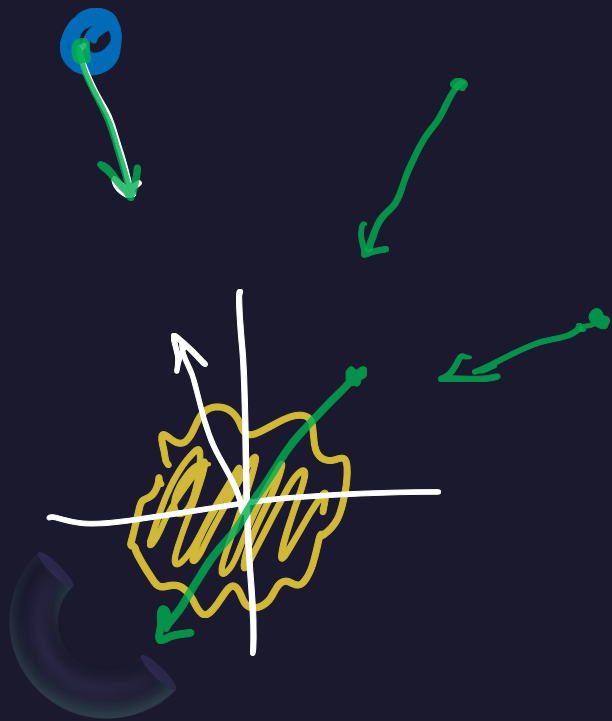


$$|F(x, y)| = \sqrt{(-y)^2 + x^2} = \sqrt{x^2 + y^2} = |(x, y)|$$

Example: Sketch a vector field on the plane given by

$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$


Example: Sketch the gravitational vector field of the gravity force exerted on a planet of mass m by a star of mass M (located at the origin).
(Hint: recall that Newton's Laws of Gravitation say that the magnitude of the force is given by mMG/r^2 .)



$$\mathbf{r} = \langle x, y, z \rangle$$

$$\frac{\mathbf{r}}{|\mathbf{r}|} = \frac{\langle x, y, z \rangle}{\sqrt{x^2 + y^2 + z^2}}$$

unit vector FROM Sun
to planet!

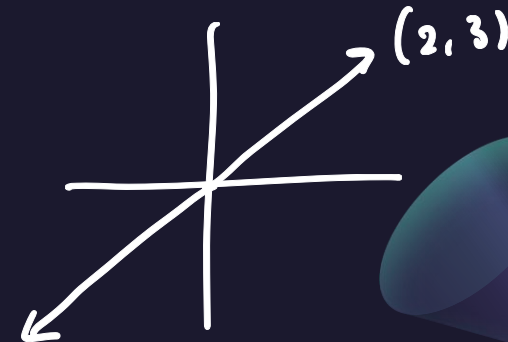
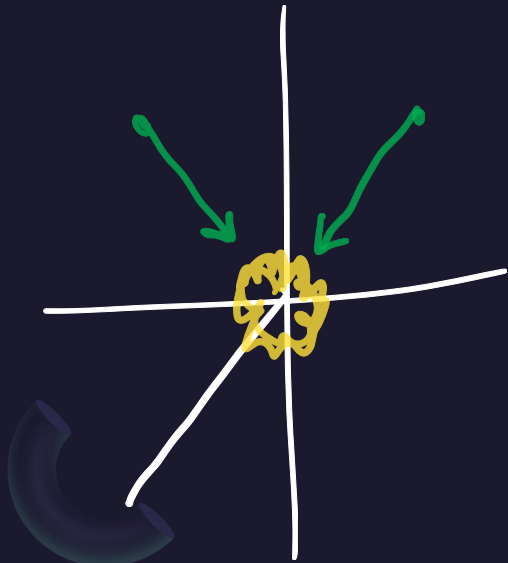
$$-\frac{\mathbf{r}}{|\mathbf{r}|} = -\frac{\langle x, y, z \rangle}{\sqrt{x^2 + y^2 + z^2}}$$

unit vector TO the Sun
from planet

Example: Sketch the gravitational vector field of the gravity force exerted on a planet of mass m by a star of mass M (located at the origin).

$$F(x, y, z) = \frac{m M G}{|r|^2} \cdot \left(-\frac{r}{|r|} \right) = -\frac{m M G}{|r|^3} \cdot r$$

$$= -\frac{m M G}{(x^2 + y^2 + z^2)^{3/2}} \cdot (x, y, z)$$



$$(-2, -3) = -(2, 3)$$

The Gradient Vector Field

$$f: \mathbb{R}^2 \longrightarrow \mathbb{R}$$

If f is a function of two variables x and y , then the **gradient** of f is the vector function ∇f defined by

$$\nabla f(x, y) := \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

$$\nabla f(x, y) = f_x(x, y) \mathbf{i} + f_y(x, y) \mathbf{j}$$

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}$$

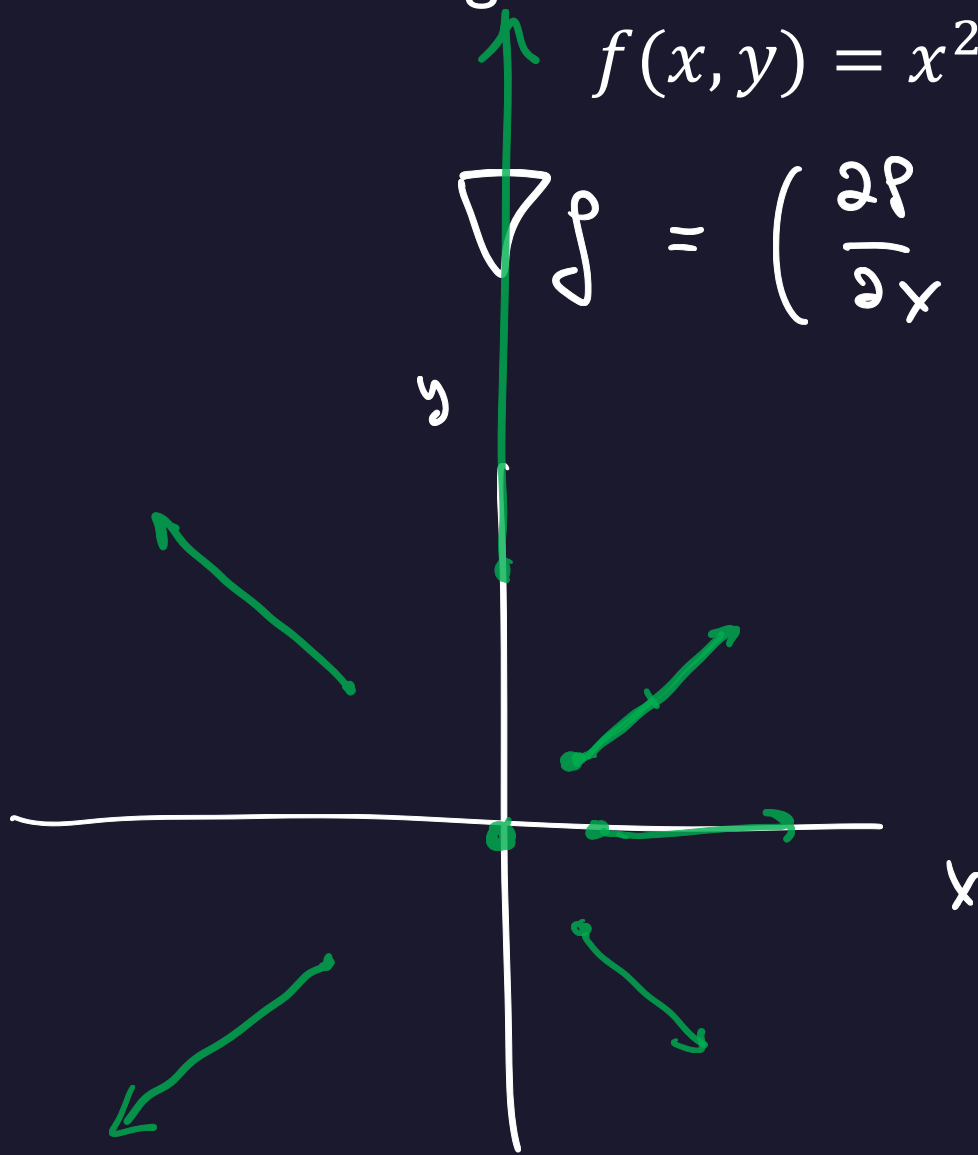
$$\nabla f(x, y, z) = f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k}$$

Example: Sketch the gradient vector field on the plane for the function

$$f(x, y) = x^2 + y^2$$

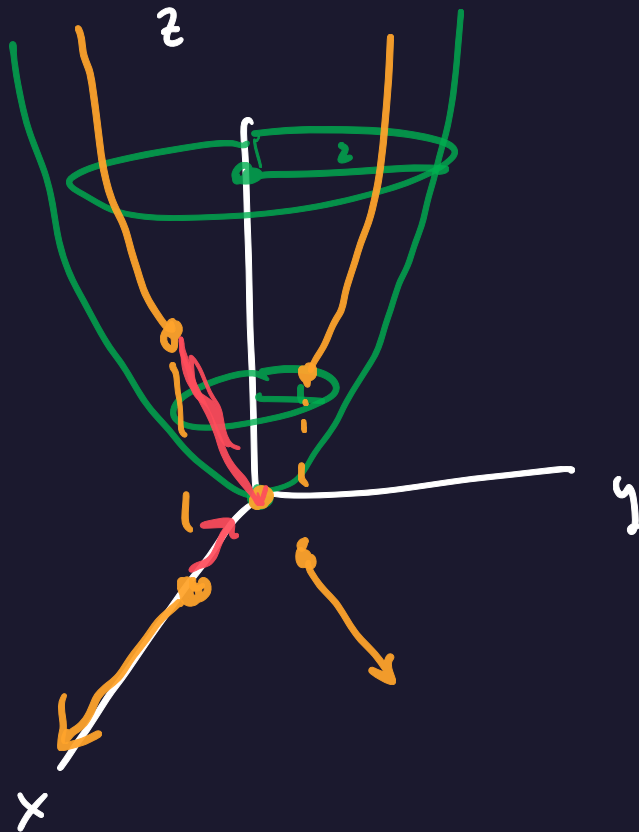
$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x, 2y) \\ = 2 \cdot (x, y)$$

$$|\nabla f| = \sqrt{(2x)^2 + (2y)^2} \\ = \sqrt{4x^2 + 4y^2} = 2\sqrt{x^2 + y^2}$$



Example: Sketch the gradient vector field on the plane for the function

$$f(x, y) = x^2 + y^2$$

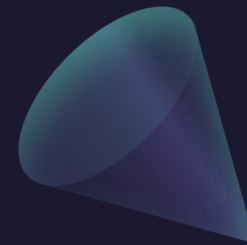


$$z = 0 = x^2 + y^2$$

$$z = 1 = x^2 + y^2$$

$$z = 4 = x^2 + y^2$$

"22"



Example: Let $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x}$ be a gravitational vector field (in 3D), and let $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$. Show that $F = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$

$$f(x, y, z) = mMG \cdot \frac{1}{\sqrt{x^2 + y^2 + z^2}} = mMG \cdot (x^2 + y^2 + z^2)^{-1/2}$$

$$\frac{\partial f}{\partial x} = mMG \cdot \left(-\frac{1}{2}\right) \cdot (x^2 + y^2 + z^2)^{-3/2} \cdot 2x = -mMG \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\frac{\partial f}{\partial y} = -mMG \cdot \frac{y}{(x^2 + y^2 + z^2)^{3/2}}, \quad \frac{\partial f}{\partial z} = -mMG \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$F(\vec{x}) = -\frac{mMG}{(x^2 + y^2 + z^2)^{3/2}} \cdot (x, y, z)$$

Example: Let $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x}$ be a gravitational vector field (in 3D), and let $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$. Show that $F = \nabla f$. ← "CONSERVATIVE VECTOR FIELD"

The Gradient Vector Field

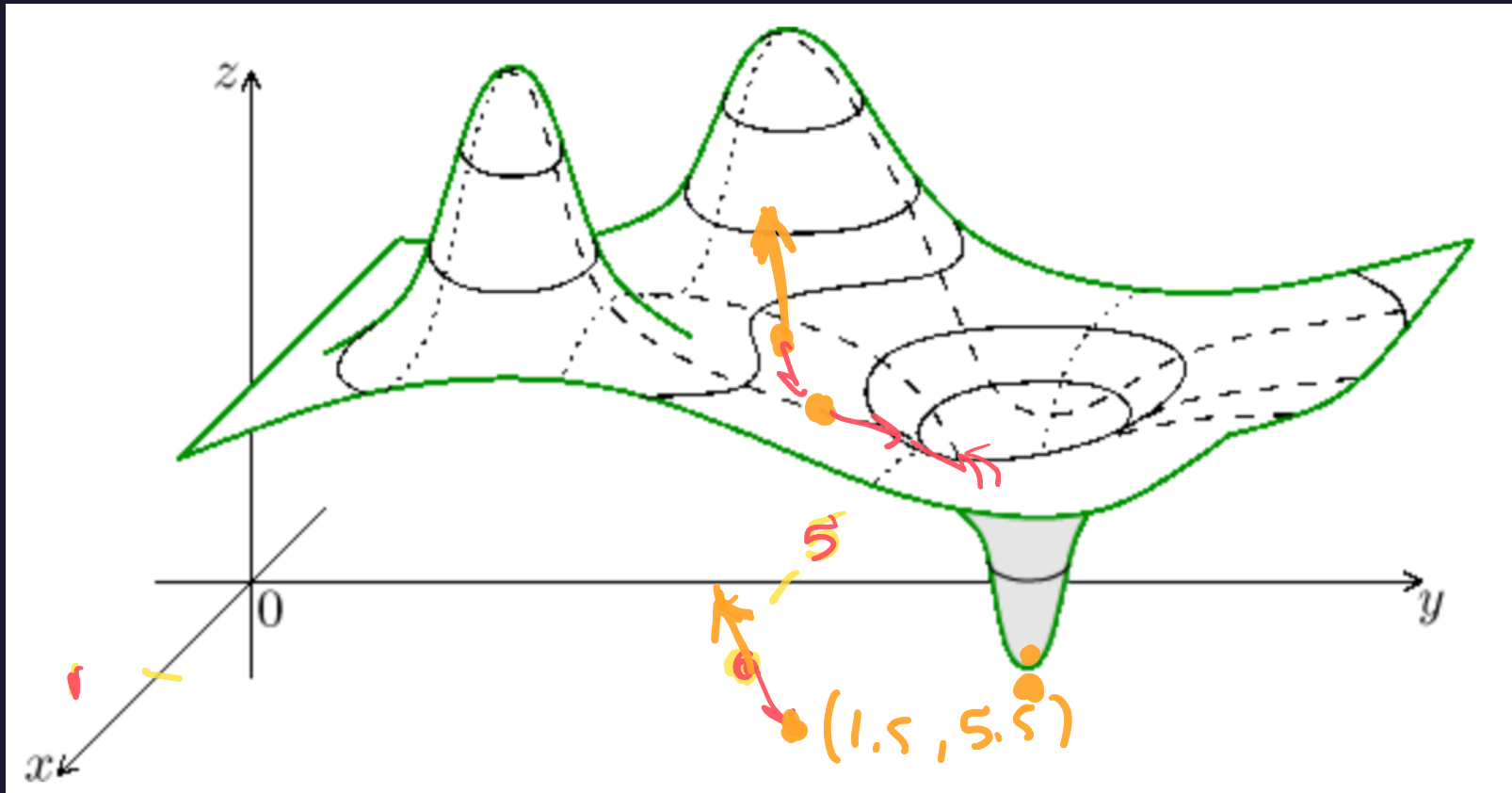
If f is a function of two variables x and y , then the **gradient** of f is the vector function ∇f defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

15 Theorem

Suppose f is a differentiable function of two or three variables. The maximum value of the directional derivative $D_{\mathbf{u}} f(\mathbf{x})$ is $|\nabla f(\mathbf{x})|$ and it occurs when $\mathbf{u}$ has the same direction as the gradient vector $\nabla f(\mathbf{x})$.

The Gradient Vector Field and “Gradient Descent” on “loss functions” in the training of AI/LLM models



Line Integrals of Vector Fields: An Example

Recall that the “work” done by a constant force F (in vector form) in moving an object from point P to point Q in space is given by

$$W = F \cdot D$$

where $D = \overrightarrow{PQ}$.

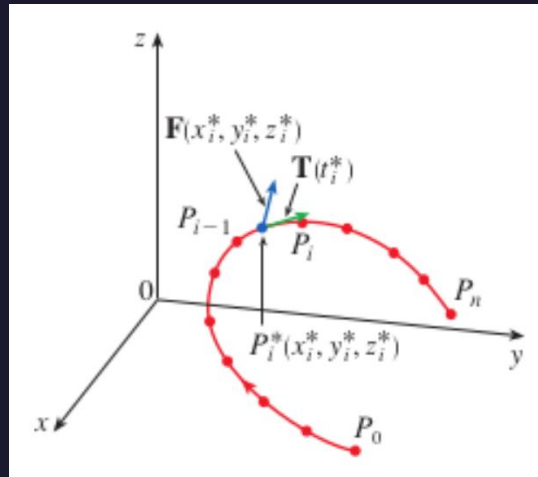
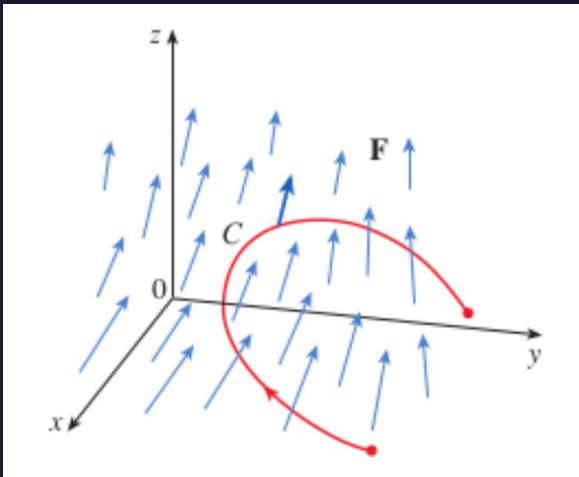
DOT
PRODUCT.

or $W = \int_a^b F(t) dt$



Line Integrals of Vector Fields: An Example

What is the “work” done by a force $\mathbf{F}$ (in vector form) in moving an object from point P to point Q in space as we travel over a curve $\mathbf{r}(t)$?

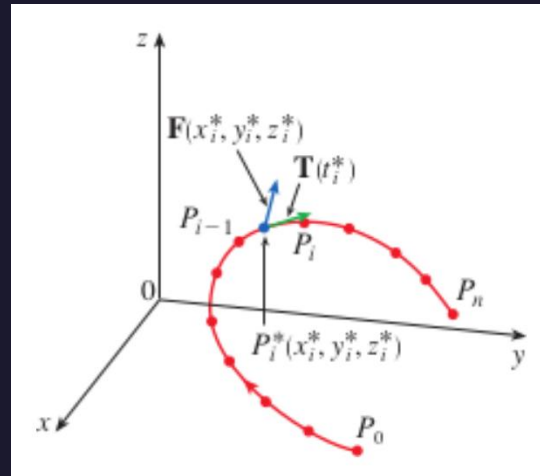
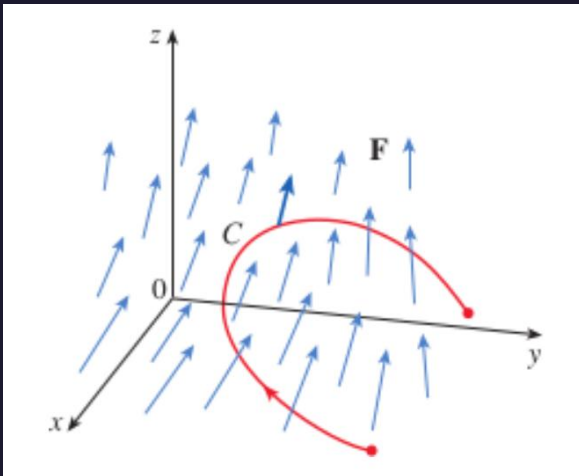


$$\sum_{i=1}^n [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i$$

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

Line Integrals of Vector Fields: An Example

What is the “work” done by a force $\mathbf{F}$ (in vector form) in moving an object from point P to point Q in space as we travel over a curve $\mathbf{r}(t)$?



$$\sum_{i=1}^n [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i$$

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

$$W = \int_a^b \left[\mathbf{F}(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \right] |\mathbf{r}'(t)| dt = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Example: Find the work done by the force

$$F(x, y) = x^2 \cdot i - xy \cdot j$$

in moving a particle along the quarter circle from (1,0) to (0,1).

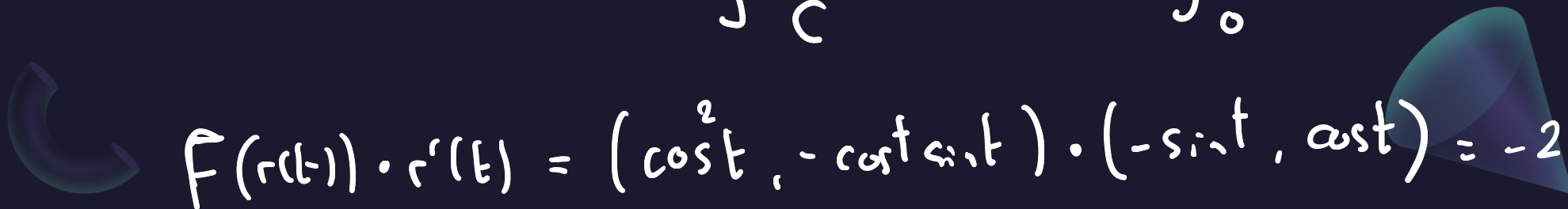
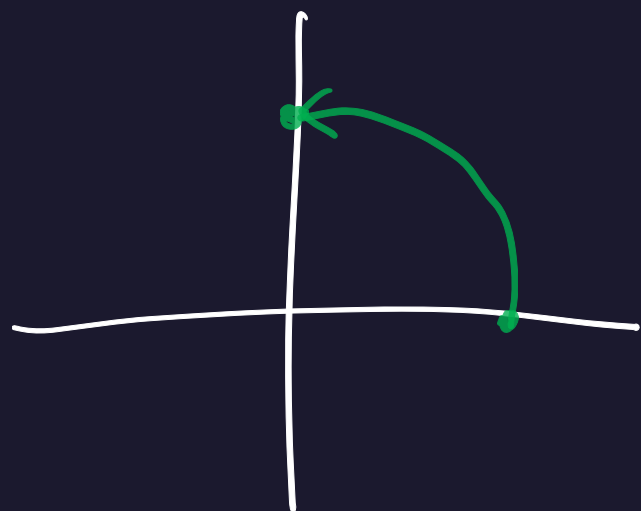
$$r(t) = (\cos t, \sin t) \quad 0 \leq t \leq \pi/2$$

$$r'(t) = (-\sin t, \cos t)$$

$$F(r(t)) = (\cos^2 t, -\cos t \sin t)$$

$$W = \int_C F \cdot dr = \int_0^{\pi/2} F(r(t)) \cdot r'(t) dt$$

$$F(r(t)) \cdot r'(t) = (\cos^2 t, -\cos t \sin t) \cdot (-\sin t, \cos t) = -2 \cos^2 t \sin t$$



Example: Find the work done by the force

$$F(x, y) = x^2 \cdot i - xy \cdot j$$

in moving a particle along the quarter circle from (1,0) to (0,1).

$$\begin{aligned} W &= \int_0^{\pi/2} F(r(t)) \cdot r'(t) \, dt \\ &= \int_0^{\pi/2} -2 \cos^2 t \sin t \, dt = \dots = \boxed{-\frac{2}{3}} \end{aligned}$$

$u = \cos t$
 $du = -\sin t \, dt$

“Work” represents the amount of energy necessary to move an object in a particular path...



Example: Set up an integral that describes the work done by gravitational forces on Artemis II during its path $r(t)$.

$$F(r) = \frac{GmM_E}{|0 - r(t)|^3} \cdot (\vec{0} - r(t)) \\ + \frac{GmM_L}{|L(t) - r(t)|^3} \cdot (L(t) - r(t)) + \frac{GmM_S}{|S(t) - r(t)|^3} \cdot (S(t) - r(t))$$

$$W = \int_0^{10} F(r(t)) \cdot r'(t) dt$$

Line Integrals of Vector Fields in General

Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve C given by a vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Then the **line integral of $\mathbf{F}$ along C** is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

Example: Evaluate $\int_C F \cdot dr$ where

$$F(x, y, z) = xy \cdot i + yz \cdot j + xz \cdot k$$

and C is the “twisted cubic” given by $r(t) = (t, t^2, t^3)$, $0 \leq t \leq 1$.

$$r'(t) = (1, 2t, 3t^2)$$

$$F(r(t)) = (t^3, t^5, t^4)$$

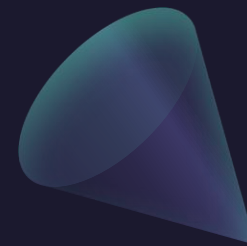
$$F(r(t)) \cdot r'(t) = 1 \cdot t^3 + 2t \cdot t^5 + 3t^2 \cdot t^4 = t^3 + 5t^6$$

$$\int_C F \cdot dr = \int_0^1 t^3 + 5t^6 dt = \boxed{\frac{27}{28}}$$

Example: Evaluate $\int_C F \cdot dr$ where

$$F(x, y, z) = xy \cdot i + yz \cdot j + xz \cdot k$$

and C is the “twisted cubic” given by $r(t) = (t, t^2, t^3)$, $0 \leq t \leq 1$.



Line Integrals of Vector Fields in General

Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve C given by a vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Then the **line integral of $\mathbf{F}$ along C** is

$$\int_a^b P(\mathbf{r}(t)) \cdot x'(t) + Q(\mathbf{r}(t)) \cdot y'(t) + R \cdot z'(t) dt$$
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

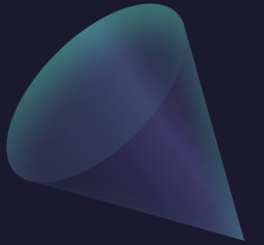
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

$$\int_a^b P(x(t), y(t), z(t)) \cdot x'(t) dt$$

Line Integrals of Vector Fields in General

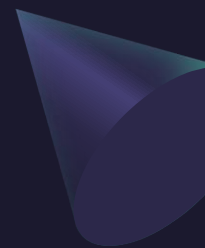
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P \, dx + Q \, dy + R \, dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

Questions?





ALVARO: Start the recording!



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Multi-Variable Calculus

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The Fundamental Theorem for Line Integrals



The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function F such that $F' = f$.

$$\int_a^b F'(x) \, dx = F(b) - F(a)$$

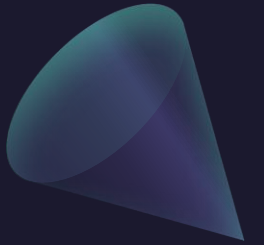
The Fundamental Theorem for Line Integrals

2 Theorem

Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

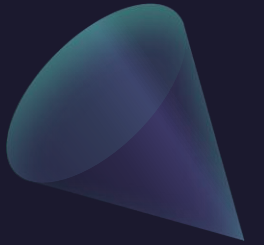
The Fundamental Theorem for Line Integrals: Proof



The Fundamental Theorem for Line Integrals: Proof

$$\begin{aligned}\int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\&= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\&= \int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt \quad \text{(by the Chain Rule)} \\&= f(\mathbf{r}(b)) - f(\mathbf{r}(a))\end{aligned}$$

Questions?



Thank you

Until next time.

